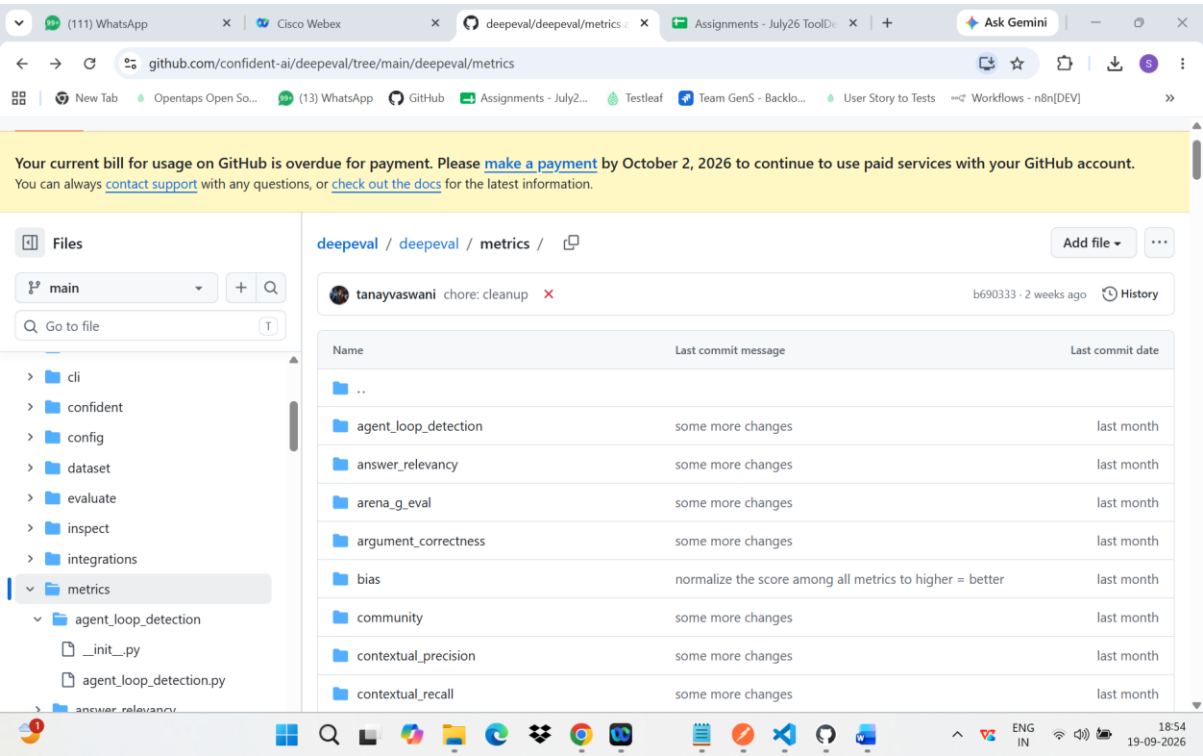


DeepEval Metrics

Explore the different evaluation metrics available in the DeepEval GitHub repository.



DeepEval’s current public metric catalog (from [deepeval.metrics](https://github.com/confident-ai/deepeval/tree/main/deepeval/metrics)):

Category	Metrics
Core / custom evaluation	GEval, ArenaGEval, ConversationalGEval, DAGMetric, ConversationalDAGMetric
Non-LLM	ExactMatchMetric, PatternMatchMetric
RAG	AnswerRelevancyMetric, FaithfulnessMetric, ContextualRecallMetric, ContextualRelevancyMetric, ContextualPrecisionMetric
MCP	MCPTaskCompletionMetric, MultiTurnMCPUseMetric, MCPUseMetric
Content quality	HallucinationMetric, BiasMetric, ToxicityMetric, SummarizationMetric
Safety / compliance	PIILeakageMetric, NonAdviceMetric, MisuseMetric, RoleViolationMetric, ToolPermissionMetric, RoleAdherenceMetric

Category	Metrics
Task-specific	ToolCorrectnessMetric, JsonCorrectnessMetric, PromptAlignmentMetric, TaskCompletionMetric, ArgumentCorrectnessMetric, KnowledgeRetentionMetric
Agentic	TopicAdherenceMetric, StepEfficiencyMetric, PlanAdherenceMetric, PlanQualityMetric, ToolUseMetric, GoalAccuracyMetric, AgentLoopDetectionMetric
Conversation / turn-level	TurnRelevancyMetric, ConversationCompletenessMetric, TurnFaithfulnessMetric, TurnContextualPrecisionMetric, TurnContextualRecallMetric, TurnContextualRelevancyMetric
Multimodal image	TextToImageMetric, ImageEditingMetric, ImageCoherenceMetric, ImageHelpfulnessMetric, ImageReferenceMetric
Voice	VoiceNaturalnessMetric, TurnTakingNaturalnessMetric, SpeechIntelligibilityMetric, VoiceConsistencyMetric, AgentResponsivenessMetric, AudioIntegrityMetric, VoiceReliabilityMetric

There are **57 concrete public metrics/evaluators** above, excluding base classes (BaseMetric, BaseConversationalMetric, BaseArenaMetric) and helper types such as GEvalTemplate and DeepAcyclicGraph.

Practise the following metrics using multiple examples:

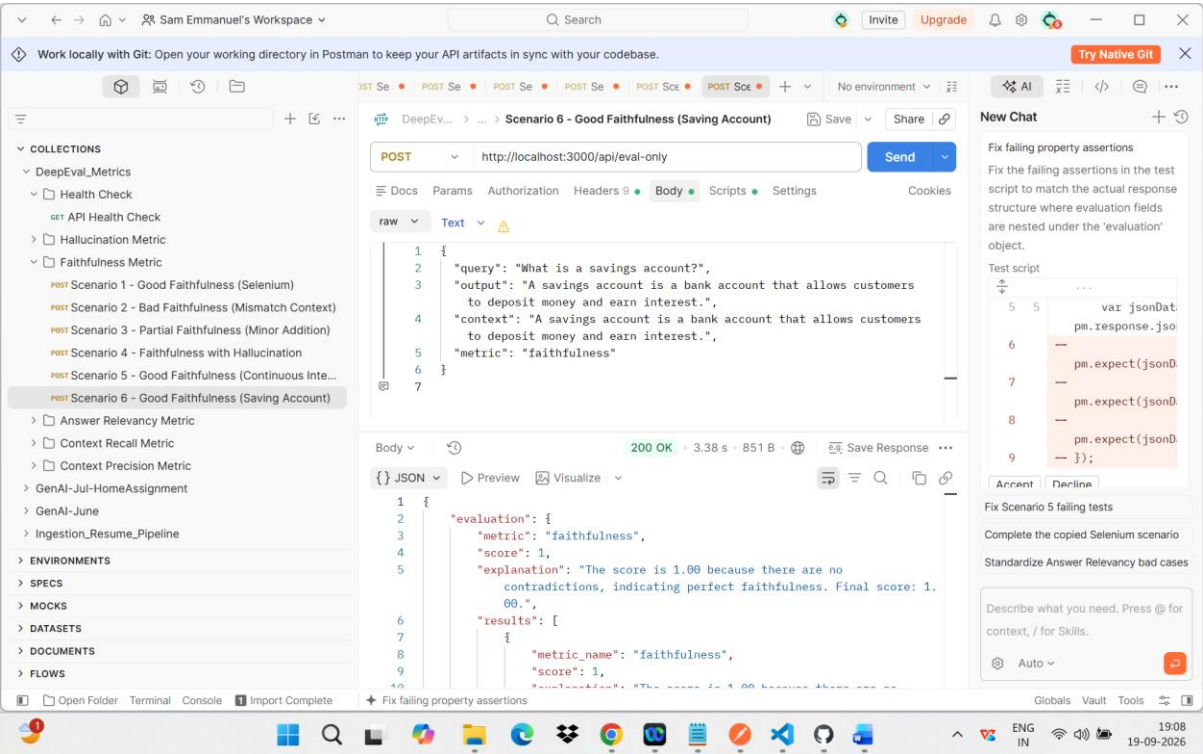
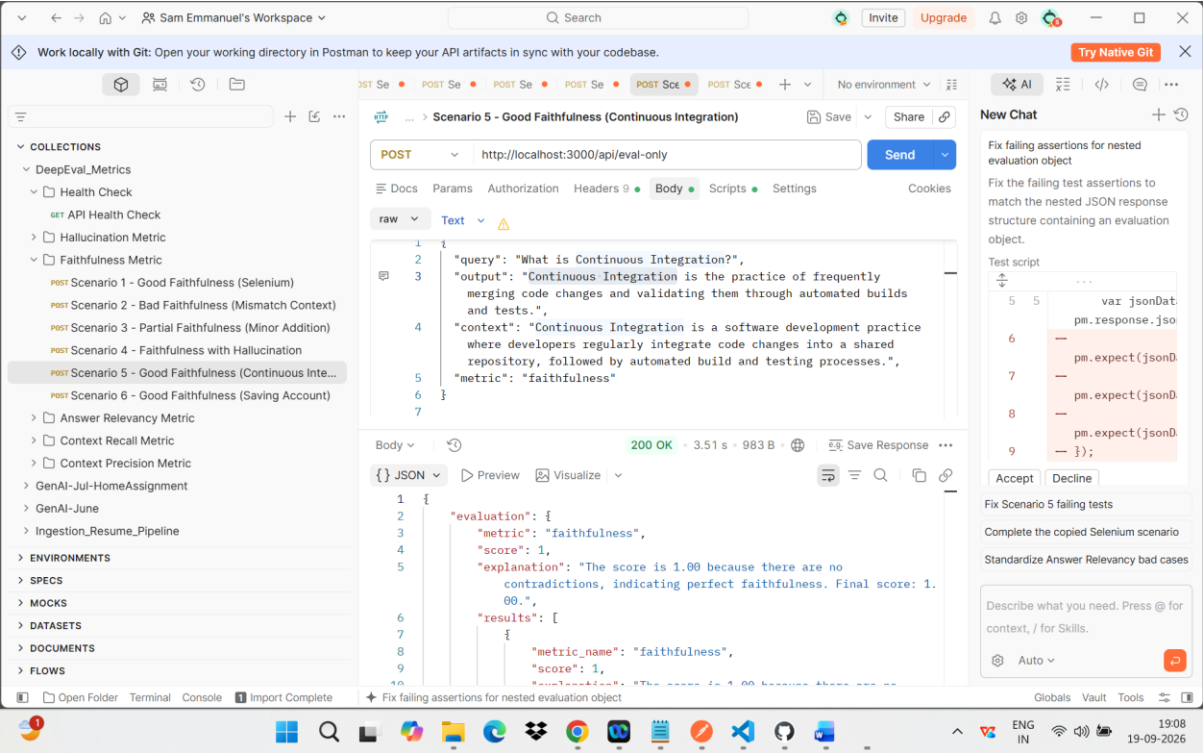
1. Faithfulness
2. Answer Relevancy

For each metric, create and test different examples that demonstrate:

- A good response with a high score
- A partially correct response with a medium score
- A poor response with a low score

Faithfulness Metric

Good Faithfulness



Partial Faithfulness

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase. [Try Native Git](#)

Scenario 7 - Partial Faithfulness (test cases for login)

POST [Send](#)

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

raw Text

```
1 {
2   "query": "Generate test cases for login.",
3   "context": [
4     "The application supports username and password authentication only."
5   ],
6   "output": "1. Verify login using username and password. 2. Verify biometric login. Biometric login is the only supported authentication method.",
7   "metric": "faithfulness"
8 }
9
```

Body JSON Preview Visualize

```
1 {
2   "evaluation": {
3     "metric": "faithfulness",
4     "score": 0.6666666666666666,
5     "explanation": "The score is 0.67 because the actual output incorrectly claims biometric login is the only supported method, contradicting the retrieval context which states the application supports only username and password authentication. Final score: 0.67.",
6     "results": [
7       {
8         "metric": "faithfulness",
9         "score": 0.67,
10        "explanation": "The score is 0.67 because the actual output incorrectly claims biometric login is the only supported method, contradicting the retrieval context which states the application supports only username and password authentication. Final score: 0.67."
11      }
12    ]
13  }
14 }
```

200 OK · 5.07 s · 1.15 KB

Fix the evaluation score assertion

Fix the failing property assertion to check jsonData.evaluation.score instead of the root object.

Test script

```
5 5 var jsonData
6 pm.response.json()
7 6 pm.expect(jsonData.evaluation.score).to.be.closeTo(0.67, 0.01);
7 7
```

Accept Decline

Fix Scenario 5 failing tests

Complete the copied Selenium scenario

Standardize Answer Relevancy bad cases

Describe what you need. Press @ for context, / for Skills.

Auto

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase. [Try Native Git](#)

Scenario 8 - Partial Faithfulness (fund transfer work)

POST [Send](#)

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

raw Text

```
1 {
2   "query": "How does fund transfer work?",
3   "context": [
4     "Users must complete OTP verification before transferring funds."
5   ],
6   "output": "Users must complete OTP verification before transferring funds. OTP verification is not required for fund transfers.",
7   "metric": "faithfulness"
8 }
9
```

Body JSON Preview Visualize

```
1 {
2   "evaluation": {
3     "metric": "faithfulness",
4     "score": 0.5,
5     "explanation": "The score is 0.50 because the actual output incorrectly states that OTP verification is not required for transfers, directly contradicting the context which mandates completing OTP verification before transferring funds. Final score: 0.50.",
6     "results": [
7       {
8         "metric": "faithfulness",
9         "score": 0.5,
10        "explanation": "The score is 0.50 because the actual output incorrectly states that OTP verification is not required for transfers, directly contradicting the context which mandates completing OTP verification before transferring funds. Final score: 0.50."
11      }
12    ]
13  }
14 }
```

200 OK · 3.73 s · 1.09 KB

Fix the evaluation score assertion

Fix the failing assertion to check for the 'score' property inside the nested 'evaluation' object.

Test script

```
5 5 var jsonData
6 pm.response.json()
7 6 pm.expect(jsonData.evaluation.score).to.be.closeTo(0.5, 0.01);
7 7
```

Accept Decline

Start using Agent Model!

Fix Scenario 7 test assertion

Parameterize login test generation

Unlink or relink scenario metadata

Describe what you need. Press @ for context, / for Skills.

Auto

Bad Faithfulness

Sam Emmanuel's Workspace

Search

Invite Upgrade

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase.

Try Native Git

DeepEval_Metr... > ... > Scenario 9 - Bad Faithfulness (Playwright)

POST http://localhost:3000/api/eval-only

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

raw Text

```
1 {
2   "query": "What is Playwright?",
3   "output": "Playwright is a browser automation framework developed by
4     Microsoft.",
5   "context": "Playwright is a stage performance written for theatrical
6     production.",
7   "metric": "faithfulness"
8 }
```

Body JSON Preview Visualize

```
1 {
2   "evaluation": {
3     "metric": "faithfulness",
4     "score": 0,
5     "explanation": "The score is 0.00 because the actual output
6       describes Playwright as a Microsoft browser automation
7       framework, which directly contradicts the retrieval context
8       that defines Playwright as a stage performance for theatrical
9       production. Final score: 0.00."
10  },
11   "results": [
12     {
13       "score": 0.00,
14       "explanation": "The score is 0.00 because the actual output
15         describes Playwright as a Microsoft browser automation
16         framework, which directly contradicts the retrieval context
17         that defines Playwright as a stage performance for theatrical
18         production. Final score: 0.00."
19     }
20   ]
21 }
```

200 OK · 3.60 s · 1.06 KB

Fix the score test assertion

New Chat

Fix the score test assertion

Correct the failing assertion to reference jsonData.evaluation.score instead of jsonData.score.

Test script

```
5 5 var jsonData
6 6 pm.response.json
7 7 pm.expect(jsonData
8 8 pm.expect(jsonData
9 9 });
```

Accept Decline

Think Harder

Start using Agent Model!

Save body & rerun scenario 9

Add negative-case assertion

Propagate to other faithfulness scenarios

Describe what you need. Press @ for context, / for Skills.

Auto

Sam Emmanuel's Workspace

Search

Invite Upgrade

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase.

Try Native Git

DeepEval_Metrics > ... > Scenario 10 - Bad Faithfulness (Chrome)

POST http://localhost:3000/api/eval-only

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

raw Text

```
1 {
2   "query": "What is Chrome?",
3   "context": [
4     "Google Chrome is a web browser developed by Google."
5   ],
6   "output": "Google Chrome is a relational database management system
7     used for storing enterprise data.",
8   "metric": "faithfulness"
9 }
```

Body JSON Preview Visualize

```
1 {
2   "evaluation": {
3     "metric": "faithfulness",
4     "score": 0,
5     "explanation": "The score is 0.00 because the actual output
6       incorrectly describes Google Chrome as a relational database
7       management system for storing enterprise data, directly
8       contradicting the retrieval context that defines Chrome as a
9       web browser developed by Google. Final score: 0.00."
10  },
11   "results": [
12     {
13       "score": 0.00,
14       "explanation": "The score is 0.00 because the actual output
15         incorrectly describes Google Chrome as a relational database
16         management system for storing enterprise data, directly
17         contradicting the retrieval context that defines Chrome as a
18         web browser developed by Google. Final score: 0.00."
19     }
20   ]
21 }
```

200 OK · 4.48 s · 1.1 KB

Fix the score path

New Chat

Fix the score path in the test

Fix the test script by accessing the score from jsonData.evaluation.score instead of jsonData.score.

Test script

```
5 5 var jsonData
6 6 pm.response.json
7 7 pm.expect(jsonData
8 8 pm.expect(jsonData
9 9 });
```

Accept Decline

Think Harder

Add assertions for faithfulness

Parameterize query/context/output

Uncomment and correct the JSON body

Describe what you need. Press @ for context, / for Skills.

Auto

Answer Relevancy Metric

Good Relevancy

The screenshot shows the Postman interface with a collection named 'Answer Relevancy Metric'. The selected item is 'Se 7_Good (user has logged in successfully)'. The request is a POST to 'http://localhost:3000/api/eval-only'. The response is a 200 OK status with a JSON body:

```
{
  "query": "How do I verify that a user has logged in successfully?",
  "output": "Verify that the user is redirected to the dashboard, the correct username is displayed, and a valid authenticated session is created.",
  "metric": "answer_relevancy"
}
```

The response body is also shown in a preview view, indicating a score of 1.00 and a perfect relevancy rating.

The screenshot shows the Postman interface with a collection named 'Answer Relevancy Metric'. The selected item is 'Se 8_Good (invalid OTP)'. The request is a POST to 'http://localhost:3000/api/eval-only'. The response is a 200 OK status with a JSON body:

```
{
  "query": "What should happen when a user enters an invalid OTP?",
  "output": "The application should reject the OTP, display an appropriate error message, and allow another attempt based on the configured retry limit.",
  "metric": "answer_relevancy"
}
```

The response body is also shown in a preview view, indicating a score of 1.00 and a perfect relevancy rating.

Partial Relevancy

The screenshot shows the Postman interface with a collection named "Answer Relevancy Metric". The selected scenario is "Se 9_Partial (security testing)". The request is a POST to "http://localhost:3000/api/eval-only". The response is a JSON object with a score of 0.25. The explanation states that the score is low because the response drifted into unrelated topics like Jira test case storage and Chrome automation scripts, which are not relevant to security testing. The results array contains one entry with the metric name "answer_relevancy" and a score of 0.25.

```
POST http://localhost:3000/api/eval-only
```

```
{  "metric": "answer_relevancy",  "query": "What is security testing?",  "output": "Security testing identifies vulnerabilities and protects application data. Test cases can be stored in Jira, automation scripts can run in Chrome, and performance reports show response times."}
```

```
{  "score": 0.25,  "explanation": "The score is 0.25 because the response drifted into unrelated topics-Jira test case storage, Chrome automation scripts, and performance report details-none of which explain what security testing is, so relevance is limited.",  "results": [    {      "metric_name": "answer_relevancy",      "score": 0.25,      "explanation": "The score is 0.25 because the response drifted into unrelated topics-Jira test case storage,"    }  ]}
```

The screenshot shows the Postman interface with the same collection, but the selected scenario is "Se 10_Partial (API testing)". The request is a POST to "http://localhost:3000/api/eval-only". The response is a JSON object with a score of 0.67. The explanation states that the score is low because the response included an irrelevant comment about testing physical screen size of mobile devices, which does not pertain to API testing and thus lowered the relevance. The results array contains one entry with the metric name "answer_relevancy" and a score of 0.67.

```
POST http://localhost:3000/api/eval-only
```

```
{  "metric": "answer_relevancy",  "query": "What is API testing?",  "output": "API testing validates requests and responses between software systems. It verifies status codes and response data. It is also used to test the physical screen size of mobile devices."}
```

```
{  "evaluation": {    "metric": "answer_relevancy",    "score": 0.6666666666666666,    "explanation": "The score is 0.67 because the response included an irrelevant comment about testing physical screen size of mobile devices, which does not pertain to API testing and thus lowered the relevance.",    "results": [      {        "metric_name": "answer_relevancy",        "score": 0.6666666666666666      }    ]  }}
```

Poor Relevancy

The screenshot shows the Postman interface with a workspace named 'Sam Emmanuel's Workspace'. The left sidebar lists collections, including 'Answer Relevancy Metric' and 'Se 11_Poor (maximum character limit)'. The main panel displays a POST request to 'http://localhost:3000/api/eval-only'. The request body is a JSON object with 'query', 'output', and 'metric' fields. The response is a 200 OK status with a JSON body containing 'evaluation', 'metric', 'score', 'explanation', and 'results' fields. The 'explanation' field states: 'The score is 0.00 because the output only discussed Selenium's browser support, which is unrelated to testing a text field's maximum character limit.' The 'results' field contains an object with 'metric_name' and 'score'.

```
POST http://localhost:3000/api/eval-only
```

```
{  "query": "How do I test the maximum character limit of a text field?",  "output": "Selenium does not supports automation across Chrome, Firefox, and Edge browsers.",  "metric": "answer_relevancy"}
```

```
{  "evaluation": {    "metric": "answer_relevancy",    "score": 0,    "explanation": "The score is 0.00 because the output only discussed Selenium's browser support, which is unrelated to testing a text field's maximum character limit.",    "results": [      {        "metric_name": "answer_relevancy",        "score": 0,        "explanation": "The score is 0.00 because the output only discussed Selenium's browser support, which is unrelated to testing a text field's maximum character limit."      }    ]  }
```

The screenshot shows the Postman interface with a workspace named 'Sam Emmanuel's Workspace'. The left sidebar lists collections, including 'Answer Relevancy Metric' and 'Se 12_Poor (Sorting Table)'. The main panel displays a POST request to 'http://localhost:3000/api/eval-only'. The request body is a JSON object with 'query', 'output', and 'metric' fields. The response is a 200 OK status with a JSON body containing 'evaluation', 'metric', 'score', 'explanation', and 'results' fields. The 'explanation' field states: 'The score is 0.00 because the output only described Appium, which is unrelated to validating table sorting functionality, making the response completely irrelevant.' The 'results' field contains an object with 'metric_name' and 'score'.

```
POST http://localhost:3000/api/eval-only
```

```
{  "query": "How do I validate the sorting functionality of a table?",  "output": "Appium is an open-source framework used for automating mobile applications.",  "metric": "answer_relevancy"}
```

```
{  "evaluation": {    "metric": "answer_relevancy",    "score": 0,    "explanation": "The score is 0.00 because the output only described Appium, which is unrelated to validating table sorting functionality, making the response completely irrelevant.",    "results": [      {        "metric_name": "answer_relevancy",        "score": 0,        "explanation": "The score is 0.00 because the output only described Appium, which is unrelated to validating table sorting functionality, making the response completely irrelevant."      }    ]  }
```